

From milk to malignancy: the role of mammary stem cells in development, pregnancy and breast cancer

.

Adult stem cells of the mammary gland (MaSCs) are a highly dynamic population of cells that are responsible for the generation of the gland during puberty and its expansion during pregnancy. In recent years significant advances have been made in understanding how these cells are regulated during these developmentally important processes both in humans and in mice. Understanding how MaSCs are regulated is becoming a particularly important area of research, given that they may be particularly susceptible targets for transformation in breast cancer. Here, we summarize the identification of MaSCs, how they are regulated and the evidence for their serving as the origins of breast cancer. In particular, we focus on how changes in MaSC populations may explain both the increased risk of developing aggressive ER/PR(–) breast cancer shortly after pregnancy and the long-term decreased risk of developing ER/PR(+) tumors.